

Ex 2.1

Show that $(1, -1)$, $(1, 2)$ and $(2, 1)$ are linearly dependent.

Three vectors are said to be linearly independent iff for $\alpha v_1 + \beta v_2 + \gamma v_3 = 0$ iff $\alpha = 0, \beta = 0, \gamma = 0$. Thus, if we show that for $\alpha \neq 0, \beta \neq 0, \gamma \neq 0$, we will have shown linear dependence of v_1, v_2, v_3 . For $(2, 1), (1, 2)$ and $(1, -1)$ and $\alpha = 1, \beta = -1, \gamma = -1$:

$$1(2, 1) - 1(1, 2) - 1(1, -1) = 0$$

E 2.2

Given transformation A : $A|0\rangle \rightarrow |1\rangle, A|1\rangle \rightarrow |0\rangle$. Write matrix representation.

Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

$$A \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} a \\ c \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \Rightarrow a = 0, c = 1$$

$$A \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} b \\ d \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \Rightarrow b = 1, d = 0$$

So, $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \sigma_x$.

Now, given matrix representation of A in some other basis:

$$A|0\rangle \rightarrow \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle), \quad A|1\rangle \rightarrow \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$$A \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \Rightarrow \begin{pmatrix} a \\ c \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \Rightarrow a = \frac{1}{\sqrt{2}}, c = -\frac{1}{\sqrt{2}}$$

$$A \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \Rightarrow \begin{pmatrix} b \\ d \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \Rightarrow b = \frac{1}{\sqrt{2}}, d = \frac{1}{\sqrt{2}}$$

So $A = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$.

Bonus

i) What is A in $|+\rangle \rightarrow |+\rangle, |-\rangle \rightarrow |-\rangle$?

This is the identity transformation. The matrix for this in the Hadamard basis is the identity matrix.
 $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

ii) What is A in $|+\rangle \rightarrow |0\rangle, |-\rangle \rightarrow |1\rangle$?

This is the Hadamard transformation: $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$.

$$H|+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle$$

$$H|-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle$$

iii) What is A in $|1\rangle, |-1\rangle \rightarrow |i0\rangle, |11\rangle$? (Note: The source text for this section appears to contain mixed notation. The following is a direct transcription of the mathematical steps provided). Transform by right multiplying by the inverse of the Hadamard transformation:

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}^{-1}$$

Thus transformation A (X) has representations:

$$\begin{pmatrix} 0 & i \\ 1 & 1 \end{pmatrix}, \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ 1 & 1 \end{pmatrix}$$

Ex 2.3

Let $A : V \rightarrow W$ and $B : W \rightarrow X$ be linear operators. Show that the matrix representation for BA is the matrix product of the representations for B and A .

Sol: For $A : V \rightarrow W$, $A|v_j\rangle = \sum_i A_{\{ij\}}|w_i\rangle$. For $B : W \rightarrow X$, $B|w_i\rangle = \sum_k B_{\{ki\}}|x_k\rangle$.

$$\begin{aligned} BA|v_j\rangle &= B(A|v_j\rangle) = B\left(\sum_i A_{\{ij\}}|w_i\rangle\right) = \sum_i A_{\{ij\}}(B|w_i\rangle) \\ &= \sum_i A_{\{ij\}} \sum_k B_{\{ki\}}|x_k\rangle = \sum_k \left(\sum_i B_{\{ki\}} A_{\{ij\}}\right)|x_k\rangle \end{aligned}$$

So, $(BA)_{\{kj\}} = \sum_i B_{\{ki\}} A_{\{ij\}}$, which is the matrix product.

Ex 2.4

Show that the identity operator I has a diagonal matrix of 1s (Kronecker delta $\delta_{\{ij\}}$).

Sol: $I|v_j\rangle = \sum_i I_{\{ij\}}|v_i\rangle = |v_j\rangle$. To leave basis vectors unchanged, the i -th component of the j -th vector must be 1 if $i = j$ and 0 otherwise. Thus $I_{\{ij\}} = \delta_{\{ij\}}$.

Ex 2.5

Verify $f(y, z) = \sum_i y_i^* z_i$ is an inner product on $\mathbb{C}^n$.

Sol: i) **Conjugate Symmetry:** $\langle z | y \rangle = \sum_i z_i^* y_i = \left(\sum_i z_i y_i^*\right)^* = \langle y | z \rangle^*$. ii) **Positivity:** $\langle v | v \rangle = \sum_i |v_i|^2 \geq 0$. iii) **Linearity in 2nd argument:** $\langle y | \alpha z \rangle = \sum_i y_i^* (\alpha z_i) = \alpha \sum_i y_i^* z_i = \alpha \langle y | z \rangle$.

Ex 2.6

Show that any inner product is conjugate linear in the first argument.

Sol:

$$\langle \sum_i \alpha_i w_i | v \rangle = \langle v | \sum_i \alpha_i w_i \rangle^* = \left(\sum_i \alpha_i \langle v | w_i \rangle\right)^* = \sum_i \alpha_i^* \langle w_i | v \rangle$$

Bonus: Normalized forms For $|w\rangle = (1, 1)$, $\|w\rangle = \sqrt{1^2 + 1^2} = \sqrt{2}$. For $|v\rangle = (1, -1)$, $\|v\rangle = \sqrt{1^2 + (-1)^2} = \sqrt{2}$. Normalized: $\frac{1}{\sqrt{2}}(1, 1)$ and $\frac{1}{\sqrt{2}}(1, -1)$.

Ex 2.7

i) Verify that $|w\rangle = (1, 1)$ and $|v\rangle = (1, -1)$ are orthogonal.

Sol: If $\langle w | v \rangle = 0$, they are orthogonal.

$$\langle w | v \rangle = (1, 1) \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 1(1) + 1(-1) = 0$$

So they are orthogonal.

Ex 2.8

Prove that the Gram-Schmidt process... (Content truncated in source)

Ex 2.9

Express Pauli operators in outer product notation using basis $|0\angle, |1\angle$.

Sol:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = |0\angle\angle 1| + |1\angle\angle 0|$$

$$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = -i|0\angle\angle 1| + i|1\angle\angle 0|$$

$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = |0\angle\angle 0| - |1\angle\angle 1|$$

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = |0\angle\angle 0| + |1\angle\angle 1|$$

Ex 2.10

Suppose $|v_j\angle$ is an orthonormal basis. What is the matrix representation for $|v_j\angle\angle v_k|$?

Sol: Matrix elements $O_{\{ab\}} = \angle v_a | O | v_b \angle$. Here $O = |v_j\angle\angle v_k|$.

$$(|v_j\angle\angle v_k|)_{\{ab\}} = \angle v_a | v_j\angle\angle v_k | v_b \angle = \delta_{\{aj\}} \delta_{\{kb\}}$$

This means the matrix has a 1 at (j, k) and 0 elsewhere.

Ex 2.11

Find eigendecomposition of Pauli matrices.

i) σ_x : $\det(\sigma_x - \lambda I) = \lambda^2 - 1 = 0 \Rightarrow \lambda = \pm 1$. Eigenvectors: $|+\angle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ for $\lambda = 1$, $|-\angle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ for $\lambda = -1$.

$$\sigma_x = (+1)|+\angle\angle +| + (-1)|-\angle\angle -|$$

ii) σ_y : $\det(\sigma_y - \lambda I) = \lambda^2 - 1 = 0 \Rightarrow \lambda = \pm 1$. Eigenvectors: $|y + \angle = \frac{1}{\sqrt{2}} \begin{pmatrix} -i \\ 1 \end{pmatrix}$ for $\lambda = 1$, $|y - \angle = \frac{1}{\sqrt{2}} \begin{pmatrix} i \\ 1 \end{pmatrix}$ for $\lambda = -1$.

$$\sigma_y = (+1)|y + \angle\angle y +| + (-1)|y - \angle\angle y -|$$

iii) σ_z : $\det(\sigma_z - \lambda I) = \lambda^2 - 1 = 0 \Rightarrow \lambda = \pm 1$. Eigenvectors: $|0\angle$ for $\lambda = 1$, $|1\angle$ for $\lambda = -1$.

$$\sigma_z = (+1)|0\angle\angle 0| + (-1)|1\angle\angle 1|$$

Ex 2.12

Prove $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ is not diagonalizable.

Sol: It suffices to show it is not normal.

$$G = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad G^\dagger = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

$$G^\dagger G = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \neq \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} = GG^\dagger$$

Not normal, so not diagonalizable.

Ex 2.13

Show that $\angle w \mid v\angle^* = \angle v \mid w\angle$.

Sol: Let $|v\rangle = \sum_i a_i |i\rangle$ and $|w\rangle = \sum_j b_j |j\rangle$.

$$\langle w | v \rangle = \sum_j \sum_i b_j^* a_i \langle j | i \rangle = \sum_i b_i^* a_i$$

$$\langle w | v \rangle^* = \left(\sum_i b_i^* a_i \right)^* = \sum_i b_i a_i^*$$

$$\langle v | w \rangle = \sum_i \sum_j a_i^* b_j \langle i | j \rangle = \sum_i a_i^* b_i$$

They are equal.

Ex 2.14

Show that $\left(\sum_i a_i A_i \right)^\dagger = \sum_i a_i^* A_i^\dagger$.

Sol: Adjoint of sum is sum of adjoints: $\left(\sum_i a_i A_i \right)^\dagger = \sum_i (a_i A_i)^\dagger$. Adjoint of scalar times operator: $(a_i A_i)^\dagger = a_i^* A_i^\dagger$. Thus $\left(\sum_i a_i A_i \right)^\dagger = \sum_i a_i^* A_i^\dagger$.

Ex 2.15

Show that $(A^\dagger)^T = A^*$.

Sol: $A^\dagger = (A^T)^*$. $(A^\dagger)^T = ((A^T)^*)^T = ((A^T)^T)^* = A^*$.

Ex 2.16

Show $P^2 = P$ for $P = \sum_i |i\rangle\langle i|$.

Sol:

$$P^2 = \left(\sum_i |i\rangle\langle i| \right) \left(\sum_j |j\rangle\langle j| \right) = \sum_i \sum_j |i\rangle\langle i | j\rangle\langle j|$$

Using $\langle i | j \rangle = \delta_{ij}$:

$$P^2 = \sum_i |i\rangle\langle i| = P$$

Ex 2.17

Show that for a normal operator A , if it has real eigenvalues, then it is self-adjoint.

Sol: $A = \sum_i \lambda_i |i\rangle\langle i|$. Since λ_i are real, $\lambda_i = \lambda_i^*$.

$$A^\dagger = \sum_i \lambda_i^* |i\rangle\langle i| = \sum_i \lambda_i |i\rangle\langle i| = A$$

Ex 2.18

Show unitary matrix eigenvalues have modulus 1.

Sol: $AA^\dagger = I$. Let $A|v\rangle = \lambda|v\rangle$.

$$\|A|v\rangle\|^2 = \langle v|A^\dagger A|v\rangle = \langle v | v \rangle = \|v\|^2$$

Also $\|A|v\rangle\|^2 = |\lambda|^2 \|v\|^2$. Thus $|\lambda|^2 = 1 \Rightarrow |\lambda| = 1$.

Ex 2.19

Show $\sigma_x, \sigma_y, \sigma_z$ are self-adjoint and unitary.

Sol: For σ_z : $\sigma_z^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}^{T*} = \sigma_z$ (Self-adjoint). $\sigma_z^2 = I$ (Unitary). Same applies to σ_x and σ_y .

Ex 2.20

(Empty in source)

Ex 2.21

(Empty in source)

Ex 2.22

Show eigenvectors of Hermitian operator with different eigenvalues are orthogonal.

Sol: $A = A^\dagger$. Let $A|i\rangle = \lambda_i|i\rangle$ and $A|j\rangle = \lambda_j|j\rangle$.

$$\langle i|A|j\rangle = \lambda_j\langle i|j\rangle$$

. Also $\langle i|A|j\rangle = (A|i\rangle)^\dagger|j\rangle = \lambda_i^*\langle i|j\rangle = \lambda_i\langle i|j\rangle$ (since real eigenvalues). $(\lambda_i - \lambda_j)\langle i|j\rangle = 0$. Since $\lambda_i \neq \lambda_j$, $\langle i|j\rangle = 0$.

Ex 2.23

Problem: A projector is a normal operator.

Sol: $P^2 = P, P = P^\dagger$.

$$PP^\dagger = P^2 = P$$

.

$$P^\dagger P = P^2 = P$$

. Commutes, so normal.

Ex 2.24

Let A be positive. Show $\langle v|A|v\rangle \geq 0$.

Sol: $A = B^\dagger B$.

$$\langle v|A|v\rangle = \langle v|B^\dagger B|v\rangle = \|B|v\rangle\|^2 \geq 0$$

.

Ex 2.25

Show $A^\dagger A$ is positive.

Sol: Show $\langle v|A^\dagger A|v\rangle \geq 0$.

$$\langle v|A^\dagger A|v\rangle = \|A|v\rangle\|^2 \geq 0$$

.

Ex 2.26

Given $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. i) Calculate $|\psi\rangle^{\otimes 2}$. ii) Calculate $|\psi\rangle^{\otimes 3}$.

Sol: i) $|\psi\rangle \otimes |\psi\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$. Vector: $\frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$.

ii) $|\psi\rangle^{\otimes 2} \otimes |\psi\rangle = \frac{1}{\sqrt{8}}(|000\rangle + |001\rangle + \dots + |111\rangle)$. Vector: $\frac{1}{\sqrt{8}}(1, 1, \dots, 1)^T$.

Ex 2.27

Problem: Calculate the following tensor products: a) $X \times Z$, b) $I \times X$, c) $X \times I$.

Solution: The matrices are $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, and $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

a) The tensor product is:

$$X \otimes Z = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0Z & 1Z \\ 1Z & 0Z \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}$$

b) The tensor product is:

$$I \otimes X = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \otimes \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1X & 0X \\ 0X & 1X \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

c) The tensor product is:

$$X \otimes I = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0I & 1I \\ 1I & 0I \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

Note that $X \otimes I \neq I \otimes X$, so the tensor product is not commutative.

Ex 2.28

Show that the transpose, complex conjugate, and adjoint operations distribute over the tensor product.

i) Show that $(A \otimes B)^* = A^* \otimes B^*$.

$$(A \otimes B)^* = [A_{\{ij\}} [B_{\{ij\}}]]^* = [A_{\{ij\}}^* [B_{\{ij\}}^*]] = A^* \otimes B^*$$

ii) Show that $(A \otimes B)^T = A^T \otimes B^T$.

$$(A \otimes B)^T = [A_{\{ij\}} [B_{\{ij\}}]]^T = [A_{\{ji\}} [B_{\{ji\}}]] = [A_{\{ji\}} B^T] = A^T \otimes B^T$$

iii) Show that $(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger$.

$$(A \otimes B)^\dagger = ((A \otimes B)^T)^* = (A^T \otimes B^T)^* = (A^T)^* \otimes (B^T)^* = A^\dagger \otimes B^\dagger$$

Ex 2.29

Given that $UU^\dagger = U^\dagger U = I$ and $VV^\dagger = V^\dagger V = I$, show that $(U \otimes V)$ is unitary.

$$(U \otimes V)(U \otimes V)^\dagger = (U \otimes V)(U^\dagger \otimes V^\dagger) = UU^\dagger \otimes VV^\dagger = I \otimes I = I$$

This shows unitarity. Similarly:

$$(U \otimes V)^\dagger(U \otimes V) = (U^\dagger \otimes V^\dagger)(U \otimes V) = U^\dagger U \otimes V^\dagger V = I \otimes I = I$$

Ex 2.30

Show that the tensor product of two Hermitian operators is Hermitian. Given $A = A^\dagger$ and $B = B^\dagger$.

$$(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger = A \otimes B$$

Thus, $(A \otimes B)^\dagger = A \otimes B$, which means the tensor product is Hermitian.

Ex 2.31

Show that the tensor product of two positive operators is positive. For A with $\langle v|A|v \rangle \geq 0$ and B with $\langle w|B|w \rangle \geq 0$, show that $\langle v \otimes w|A \otimes B|v \otimes w \rangle \geq 0$.

The inner product is given by:

$$\begin{aligned}\angle v \otimes w | (A \otimes B) | v \otimes w \angle &= (\angle v | \otimes \angle w |) (A \otimes B) (| v \angle \otimes | w \angle) \\ &= \angle v | A | v \angle \angle w | B | w \angle\end{aligned}$$

Since we are given that $\angle v | A | v \angle \geq 0$ and $\angle w | B | w \angle \geq 0$, and multiplication of non-negative real numbers results in a non-negative real number:

$$\angle v | A | v \angle \angle w | B | w \angle \geq 0$$

This proves the statement.

Ex 2.32

Show that the tensor product of two projectors is a projector. Let $A = \sum_i |i \angle \angle i|$ be a projector in V , and $B = \sum_k |k \angle \angle k|$ be a projector in W .

$$\begin{aligned}(A \otimes B) &= \left(\sum_i |i \angle \angle i| \right) \otimes \left(\sum_k |k \angle \angle k| \right) = \sum_{i,k} (|i \angle \otimes |k \angle) (\angle i | \otimes \angle k |) \\ &= \sum_{i,k} |ik \angle \angle ik|\end{aligned}$$

This describes a projector in $V \otimes W$. This shows that projectors are closed under the tensor product.

Ex 2.33

Show that $H^{\otimes n} = \frac{1}{\sqrt{2}^n} \sum_{x,y=0}^{2^n-1} (-1)^{x \cdot y} |x \angle \angle y|$.

Consider $H^{\otimes 2}$:

$$\begin{aligned}&\left(\frac{1}{\sqrt{2}} (|0 \angle + |1 \angle) \angle 0| + \frac{1}{\sqrt{2}} (|0 \angle - |1 \angle) \angle 1| \right)^{\otimes 2} \\ &= \frac{1}{2} (|0 \angle \angle 0| + |1 \angle \angle 0| + |0 \angle \angle 1| + |1 \angle \angle 1|)^{\otimes 2}\end{aligned}$$

(Correction from source logic: The source expands $H^{\otimes 2}$ by terms). Looking at the pattern, the sign depends on $x_i y_i$.

$$H^{\otimes n} = \frac{1}{\sqrt{2}^n} \sum_{x,y=0}^{2^n-1} (-1)^{x_i y_i} |x \angle \angle y|$$

Show $H^{\otimes 2}$ explicitly:

$$H^{\otimes 2} = \frac{1}{2} \sum_{x,y=0}^3 (-1)^{x \cdot y} |x \angle \angle y|$$

$$|0 \angle \angle 00| + |01 \angle \angle 01| + |10 \angle \angle 10| + |11 \angle \angle 11| + |01 \angle \angle 00| + |00 \angle \angle 01| + |11 \angle \angle 10| + |10 \angle \angle 11| + |10 \angle \angle 00| + |11 \angle \angle 01| + |00 \angle \angle 10| + |01 \angle \angle 11|$$

(Note: The explicit expansion in the source text seems to list many cross terms consistent with the tensor product expansion).

Ex 2.34

Eigenvectors and Normalization Normalize the vectors corresponding to eigenvalues of $\begin{pmatrix} 4 & 3 \\ 3 & 4 \end{pmatrix}$. Eigenvalues are 1 and 7. $|1 \angle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$; $|7 \angle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

i) Square root of the matrix:

$$\begin{aligned}\sqrt{\begin{pmatrix} 4 & 3 \\ 3 & 4 \end{pmatrix}} &= \sqrt{1}|1\angle\angle 1| + \sqrt{7}|7\angle\angle 7| \\ &= 1 \cdot \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} + \sqrt{7} \cdot \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\end{aligned}$$

ii) Logarithm of the matrix:

$$\begin{aligned}\log\left(\begin{pmatrix} 4 & 3 \\ 3 & 4 \end{pmatrix}\right) &= \log_2(1)|1\angle\angle 1| + \log_2(7)|7\angle\angle 7| \\ &= 0 + \log(7) \cdot \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\end{aligned}$$

Ex 2.35

(Empty in source)

Ex 2.36

Show that the Pauli matrices except for I have trace zero.

$$\sigma_x : \text{tr} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = 0 + 0 = 0$$

$$\sigma_y : \text{tr} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = 0 + 0 = 0$$

$$\sigma_z : \text{tr} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = 1 - 1 = 0$$

Ex 2.37

Show that $\text{tr}(AB) = \text{tr}(BA)$.

$$\text{tr}(AB) = \sum_i (AB)_{\{ii\}} = \sum_i \sum_j A_{\{ij\}} B_{\{ji\}}$$

$$\text{tr}(BA) = \sum_i (BA)_{\{ii\}} = \sum_i \sum_j B_{\{ij\}} A_{\{ji\}}$$

Since i and j are summation variables over the same index set, the two traces are equal.

Ex 2.38

i) Show that $\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B)$.

$$\text{tr}(A + B) = \sum_i (A + B)_{\{ii\}} = \sum_i (A_{\{ii\}} + B_{\{ii\}}) = \sum_i A_{\{ii\}} + \sum_i B_{\{ii\}} = \text{tr}(A) + \text{tr}(B)$$

ii) Show that $\text{tr}(zA) = z \text{tr}(A)$.

$$\text{tr}(zA) = \sum_i (zA)_{\{ii\}} = \sum_i zA_{\{ii\}} = z \sum_i A_{\{ii\}} = z \text{tr}(A)$$

Ex 2.39

Inner product of operators Show that $\angle A \mid B \angle = \text{tr}(A^\dagger B)$ satisfies inner product properties.

i) **Conjugate symmetry:**

$$\angle A \mid B \angle^* = (\text{tr}(A^\dagger B))^* = \text{tr}((A^\dagger B)^\dagger) = \text{tr}(B^\dagger A) = \angle B \mid A \angle$$

ii) **Linearity in second argument:**

$$\angle A \mid \alpha B + \beta C \angle = \text{tr}(A^\dagger(\alpha B + \beta C)) = \alpha \text{tr}(A^\dagger B) + \beta \text{tr}(A^\dagger C) = \alpha \angle A \mid B \angle + \beta \angle A \mid C \angle$$

iii) **Positive definiteness:**

$$\angle A \mid A \angle = \text{tr}(A^\dagger A) = \sum_i (A^\dagger A)_{\{ii\}} = \sum_i \sum_j (A^\dagger)_{\{ij\}} A_{\{ji\}} = \sum_{i,j} A_{\{ji\}}^* A_{\{ji\}} = \sum_{i,j} |A_{\{ji\}}|^2$$

This sum of squares is always non-negative and is zero iff all $A_{\{ji\}} = 0$, i.e., $A = 0$.

Ex 2.40

Verify that $[X, Y] = 2iZ$ for Pauli matrices.

$$\begin{aligned} XY - YX &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} - \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} - \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix} \\ &= \begin{pmatrix} 2i & 0 \\ 0 & -2i \end{pmatrix} = 2i \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = 2iZ \end{aligned}$$

Ex 2.41

Verify that $\sigma_i \sigma_j + \sigma_j \sigma_i = 2\delta_{\{ij\}} I$.

For $i = j$: $\sigma_i^2 = I$.

$$\begin{aligned} \sigma_x^2 &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \\ \sigma_y^2 &= \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \\ \sigma_z^2 &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \end{aligned}$$

For $i \neq j$: $\sigma_i \sigma_j + \sigma_j \sigma_i = 0$.

Examples:

- For $i = 1, j = 2$ (X, Y):

$$XY + YX = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} + \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = 0$$

- For $i = 2, j = 3$ (Y, Z):

$$YZ + ZY = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} + \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix} = 0$$

- For $i = 1, j = 3$ (X, Z):

$$XZ + ZX = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = 0$$

Ex 2.42

Verify that $AB = \frac{1}{2}([A, B] + \{A, B\})$.

Consider the RHS:

$$\frac{1}{2}([A, B] + \{A, B\}) = \frac{1}{2}((AB - BA) + (AB + BA)) = \frac{1}{2}(2AB) = AB$$

Ex 2.43

Show that for $j, k = 1, 2, 3$,

$$\sigma_j \sigma_k = \delta_{\{jk\}} I + i \sum_{\{l=1\}}^3 \varepsilon_{\{jkl\}} \sigma_l$$

a) For $j = k = 1$:

$$\sigma_1 \sigma_1 = I$$

. RHS: $\delta_{\{11\}} I + i \sum \varepsilon_{\{11l\}} \sigma_l = I + 0 = I$. (Holds)

b) For $j = 1, k = 2$:

$$\sigma_1 \sigma_2 = \sigma_x \sigma_y = i \sigma_z$$

. RHS: $\delta_{\{12\}} I + i \sum \varepsilon_{\{12l\}} \sigma_l = 0 + i(\varepsilon_{\{123\}} \sigma_3) = i \sigma_z$. (Holds)

c) For $j = 1, k = 3$:

$$\sigma_1 \sigma_3 = \sigma_x \sigma_z = -i \sigma_y$$

. RHS: $\delta_{\{13\}} I + i \sum \varepsilon_{\{13l\}} \sigma_l = 0 + i(\varepsilon_{\{132\}} \sigma_2) = i(-1) \sigma_2 = -i \sigma_y$. (Holds)

Ex 2.44

If $[A, B] = 0$ and $\{A, B\} = 0$, show that $B = 0$ if $A \neq 0$ (and A is invertible).

$$AB - BA = 0 \Rightarrow AB = BA$$

$$AB + BA = 0 \Rightarrow AB = -BA$$

Combining: $AB = -AB \Rightarrow 2AB = 0 \Rightarrow AB = 0$. If A is invertible, multiply by A^{-1} :

$$A^{-1}AB = A^{-1}0 \Rightarrow B = 0$$

.

Ex 2.45

Show that $[A, B]^\dagger = [B^\dagger, A^\dagger]$.

Proof:

$$[A, B]^\dagger = (AB - BA)^\dagger = (AB)^\dagger - (BA)^\dagger = B^\dagger A^\dagger - A^\dagger B^\dagger = [B^\dagger, A^\dagger]$$

Ex 2.46

Show that $[A, B] = -[B, A]$.

Proof:

$$[A, B] = AB - BA = -(BA - AB) = -[B, A]$$

Ex 2.47

For $A = A^\dagger$ and $B = B^\dagger$, show that $i[A, B]$ is Hermitian.

$$(i[A, B])^\dagger = i^\dagger [A, B]^\dagger = -i [B^\dagger, A^\dagger] = -i[B, A]$$

Using $[B, A] = -[A, B]$:

$$= -i(-[A, B]) = i[A, B]$$

Thus, it is Hermitian.

Ex 2.48

Polar decomposition of a normal matrix. Let A be a normal matrix ($AA^\dagger = A^\dagger A$).

$$\begin{aligned} A &= PU = (AA^\dagger)^{0.5}U = (A^\dagger A)^{0.5}U = UP \\ &\Rightarrow A = PU = UP \end{aligned}$$

Where U is unitary ($UU^\dagger = U^\dagger U = I$) and P is positive ($\langle v | Pv \rangle \geq 0$).

Ex 2.49

a) If $A = PU$ and let positive operator P be A .

$$\begin{aligned} \Rightarrow A &= AU \Rightarrow U = I \\ \Rightarrow A &= AI \end{aligned}$$

(Right polar decomposition)

$$\Rightarrow A = IA$$

(Left polar decomposition)

b) Find $PU = A$ for unitary U ($AA^\dagger = A^\dagger A = I$).

$$\begin{aligned} A &= \sqrt{AA^\dagger}U = \sqrt{I}U = IU \\ A &= U\sqrt{A^\dagger A} = U\sqrt{I} = UI \\ \Rightarrow A &= IU = UI \end{aligned}$$

, where I is the positive operator.

c) If $A = A^\dagger$ (Hermitian), find P and U .

$$\begin{aligned} A &= \sqrt{AA^\dagger}U = \sqrt{A^2}U = \pm AU \\ A &= U\sqrt{A^\dagger A} = U\sqrt{A^2} = U(\pm A) \end{aligned}$$

(Note: The source text ends with a contradiction regarding all Hermitian operators being positive, seemingly highlighting a nuance in the definition of $\sqrt{A^2}$ versus A itself for non-positive Hermitian operators).

Ex 2.50

Polar Decomposition Example

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

A is positive (assuming real entries and checking specific criteria, though strict positivity requires checking eigenvalues or quadratic forms). If A is positive: $A = AU \Rightarrow U = I$. Left polar decomposition:

$$\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Right polar decomposition:

$$\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

Ex 2.51

Verify that $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ is unitary.

Sol: To show see: $HH^\dagger = H^\dagger H = I$.

$$HH^\dagger = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = I$$

Ex 2.52

Verify that $H^2 = I$.

Sol:

$$H^2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = I$$

Ex 2.53

Find eigenvalues and eigenvectors of H .

Sol: $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$. Consider $\det(H - \lambda I) = 0$. Note: We find eigenvalues of the matrix part first, then scale.

$$\det \begin{pmatrix} 1 - \lambda & 1 \\ 1 & -1 - \lambda \end{pmatrix} = 0$$

$$\Rightarrow (1 - \lambda)(-1 - \lambda) - 1 = 0$$

$$\Rightarrow -1 - \lambda + \lambda + \lambda^2 - 1 = 0$$

$$\Rightarrow \lambda^2 = 2 \Rightarrow \lambda = \pm\sqrt{2}$$

Scaling by factor $\frac{1}{\sqrt{2}}$, H has eigenvalues ± 1 .

Eigenvectors of H (Detailed Calculation): i) For eigenvalue +1:

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \sqrt{1} \begin{pmatrix} a \\ b \end{pmatrix}$$

$$\frac{1}{\sqrt{2}}(a + b) = a$$

$$\frac{1}{\sqrt{2}}(a - b) = b$$

From first eq: $a + b = \sqrt{2}a \Rightarrow b = (\sqrt{2} - 1)a$. Set $a = 1, b = \sqrt{2} - 1$.

$$|v_1\rangle = \begin{pmatrix} 1 \\ \sqrt{2} - 1 \end{pmatrix}$$

Normalized:

$$\frac{1}{\sqrt{1^2 + (\sqrt{2} - 1)^2}} \begin{pmatrix} 1 \\ \sqrt{2} - 1 \end{pmatrix} = \frac{1}{\sqrt{1 + 2 - 2\sqrt{2} + 1}} \begin{pmatrix} 1 \\ \sqrt{2} - 1 \end{pmatrix} = \frac{1}{\sqrt{4 - 2\sqrt{2}}} \begin{pmatrix} 1 \\ \sqrt{2} - 1 \end{pmatrix}$$

ii) For eigenvalue -1:

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = -1 \begin{pmatrix} a \\ b \end{pmatrix}$$

$$\frac{1}{\sqrt{2}}(a+b) = -a \Rightarrow a+b = -\sqrt{2}a \Rightarrow b = -(1+\sqrt{2})a$$

. (Note: The source text derivation for part (ii) deviates here, verifying the steps provided in the source below): Source checks: $\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = -\sqrt{2}\begin{pmatrix} a \\ b \end{pmatrix}$ (Checks matrix part against scaled eigenvalue).

$$\Rightarrow a+b = -2a \Rightarrow b = -3a$$

. (This step in source seems distinct or assumes different scaling, transcription follows): Set $a = 1$: $|v_2\rangle = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$. Normalized: $\frac{1}{\sqrt{1^2+(-1)^2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

Ex 2.54

Given $[A, B] = 0$, i.e., $AB = BA$. Show that $e^A e^B = e^{A+B}$.

$$e^A e^B = \left(\sum_i e^{\lambda_i} |i\rangle\langle i| \right) \left(\sum_j e^{\lambda_j} |j\rangle\langle j| \right)$$

Assuming simultaneous diagonalization basis $|i\rangle$:

$$= \sum_i e^{\lambda_A} e^{\lambda_B} |i\rangle\langle i| = \sum_i e^{\lambda_A + \lambda_B} |i\rangle\langle i| = e^{A+B}$$

Ex 2.55

Show that $U(t, t_0) = e^{-iH\frac{t_2-t_1}{\hbar}}$ is unitary.

Consider $U(t_2, t_1)U^\dagger(t_2, t_1)$:

$$\begin{aligned} &= e^{-iH\frac{t_2-t_1}{\hbar}} e^{[-iH\frac{t_2-t_1}{\hbar}]^\dagger} \\ &= e^{-iH\frac{t_2-t_1}{\hbar}} e^{+iH^\dagger\frac{t_2-t_1}{\hbar}} \end{aligned}$$

Since H is Hermitian ($H = H^\dagger$) and commutes with itself:

$$= e^0 = I$$

Consider $U^\dagger(t_2, t_1)U(t_2, t_1)$:

$$= e^{iH^\dagger\frac{t_2-t_1}{\hbar}} e^{-iH\frac{t_2-t_1}{\hbar}} = e^0 = I$$

List of Problems

Before going further, let me see. What all problems need I need help with.

- Ex 2.1 to 2.7: Solved.
- **Ex 2.8: I need help.**
- Ex 2.9 to 2.19: Solved.
- **Ex 2.20: I need help.**
- **Ex 2.21: I need help.**
- Ex 2.22 to Ex 2.30: Solved.
- **Ex 2.31: I need help.**
- Ex 2.32: Solved.
- **Ex 2.33: I need help.**
- **Ex 2.34: I need help.**
- **Ex 2.35: I need help.**
- Ex 2.36: Solved.
- Ex 2.37 to Ex 2.38: Solved.
- **Ex 2.39: I need help.**
- Ex 2.40 to 2.47: Solved.
- **Ex 2.48: I need help.**

- Ex 2.49 to 2.52: Solved.
- **Ex 2.53: I need help.**
- Ex 2.54 to Ex 2.55: Solved.
- **Ex 2.56: I need help.**